

- Q.27 Discuss any two major nonlinearities with diagram?
- Q.28 What do you mean by friction?
- Q.29 Explain the concept of Fuzzy logic.
- Q.30 Explain the concept of Artificial neural network with suitable diagram.
- Q.31 Write at least five applications of Robotics.
- Q.32 Explain the principle of superposition and homogeneity.
- Q.33 Write a short note on PLC software.
- Q.34 What do you mean by LabVIEW?
- Q.35 Mention any five disadvantages of open loop control system.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Write eight differences between open loop and closed loop control system with example.
- Q.37 Describe the concept of ratio control system with suitable example.
- Q.38 Explain Artificial intelligence in detail with applications.

No. of Printed Pages : 4
Roll No.

181544/121544/031544

4th Sem / Instrumentation & Control Engg. Subject:- Advanced Control System

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which control loop is best handling error in the plant?
- a) cascade b) feedback
c) feedforward d) ratio
- Q.2 At which place we use actuators in control loop?
- a) forward path b) feedback path
c) both a and b d) none of these
- Q.3 _____ is one of the type of non-linearities
- a) Feedback b) Process
c) Friction d) All of the above
- Q.4 Following are robotic arm configuration.
- a) Cartesian b) Cylindrical
c) Spherical d) All of these
- Q.5 Application of robotics are
- a) Assembling b) Dismantling
c) Welding d) all of these

- Q.6 In open loop control system
- Feedback is absent
 - System will be in accurate
 - Feedback is present
 - none of these
- Q.7 Which of the following command is used to clear window in MATLAB
- clear
 - clc
 - close all
 - clear all
- Q.8 Full form of ANN
- Artificial Neural Network
 - Always Neural Network
 - Above Neural Network
 - None of the above
- Q.9 Which part of Robot control all the functions
- Sensor
 - Controller
 - Actuator
 - display device
- Q.10 Artificial intelligence can be used in
- Robotics
 - Process plants
 - Food processing plants
 - All of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is relay?
- Q.12 Give one application of feedback control.
- Q.13 Expand LabVIEW.
- Q.14 What is Backlash?
- Q.15 Expand AI.
- Q.16 Define Sensor.
- Q.17 Expand SCADA.
- Q.18 Give one application of control system.
- Q.19 Write any two multi-loop control system.
- Q.20 Expand MATLAB.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain cascade Control with applications.
- Q.22 Give at least five differences between linear and nonlinear control system.
- Q.23 Write Five applications of feed forward control.
- Q.24 What a short note on Multiloop control system.
- Q.25 Explain split range control system with advantage.
- Q.26 Explain Jump resonance with suitable diagram?